Genetics Revision Paper

Name: …… Class: …… Date: ……

Part 1: Key Points

1. Heredity and how crosses are described	
Inheritance	The passing of traits to the next generation (also called **heredity**). **Genetics** is the science of heredity.
True-breeding	Plants that produce offspring with the same trait when they self-fertilize, generation after generation (pure).
Self-pollination	The male and female gametes of the **same plant** combine.
Cross-pollination	The male and female gametes of **different plants** combine (naturally by wind or insects, or artificially by removing the anthers and moving pollen to the stigma).
P, F_1, F_2	**P** = parental generation. **F_1** (first filial) = offspring of the P cross. **F_2** (second filial) = offspring of the F_1 cross.
Example	True-breeding yellow × green seeds → all F_1 yellow; the green trait reappears in F_2 in a **3:1** ratio (yellow : green).

2. Genes, alleles and genotypes	
Gene / allele	A gene controls a trait. An **allele** is an alternative form of a single gene (the yellow-seed and green-seed alleles are alleles of the same gene).
Dominant	Controls the trait that appears in the F_1 generation; written as a **capital** letter (Y).
Recessive	Masked in the F_1 generation when a dominant allele is present; written as a **lowercase** letter (y).
Homozygous	Two identical alleles for a trait (YY or yy).
Heterozygous (hybrid)	Two different alleles for a trait (Yy). The dominant trait is observed and the recessive trait is masked.
Genotype	An organism's allele pair (YY, Yy or yy).
Phenotype	The observable characteristic (yellow or green). Yellow seeds can be YY or Yy; green seeds are always yy.

3. Laws of inheritance	
Law of segregation	The two alleles for each trait **separate during meiosis**, and two alleles unite at fertilization.
Law of independent assortment	Alleles are distributed to gametes **randomly**: genes on separate chromosomes sort independently, and each allele combination is equally likely.
Seven traits studied	Pod color, pod shape, seed color, seed shape, stem length, flower color and flower position.

4. Monohybrid and dihybrid crosses, and probability	
Punnett square	Keeps track of the possible genotypes of a cross. The number of squares depends on the number of allele types each parent produces.
Monohybrid cross	A cross involving hybrids for **one** trait (Yy × Yy). Genotypic ratio **1:2:1** (YY:Yy:yy); phenotypic ratio **3:1**.
Dihybrid cross	Inheritance of **two** traits at once (YyRr × YyRr). Each parent makes four gamete types (YR, Yr, yR, yr), giving 16 squares and a phenotypic ratio of **9:3:3:1**.
Probability	Inheritance works like flipping a coin (each allele has a ½ chance). Multiply the probabilities of independent events (**rule of multiplication**). Example: Tt × Tt → TT = ½ × ½ = ¼.
Real data	Real results may not match the predicted ratio exactly, but the more offspring there are, the closer they get to the Punnett square prediction.

Cross	Offspring genotypes	Phenotypic ratio
Tt × Tt	1 TT : 2 Tt : 1 tt	3 tall : 1 dwarf
Bb × bb	1 Bb : 1 bb	1 black : 1 white
YY × yy	all Yy	all yellow

5. Genetic recombination	
Definition	The new combinations of genes produced by **crossing over** and **independent assortment**. Genetic recombination increases variation.
Counting combinations	Gamete types from independent assortment = $\mathbf{2^n}$ (n = number of chromosome pairs). Possible zygote combinations = $2^n \times 2^n$. Example: 6 pairs → 64 gamete types and 4096 zygote types.
Sources of variation	Crossing over, independent assortment during meiosis, and random fertilization.

6. Gene linkage and chromosome maps	
Linked genes	Genes located close together on the **same chromosome**. They usually travel together during gamete formation and do **not** segregate independently, so linkage is an **exception** to the law of independent assortment.
Crossing over and distance	Crossing over happens **more often between genes that are far apart** than between genes that are close together.
Chromosome map	A depiction of how genes are arranged on a chromosome, built from crossing-over data. **1 map unit = 1% crossing over.**
Parental vs recombinant	For linked genes, parental allele combinations are more likely than recombinant ones.
Worked example	A–B = 10%, B–C = 5%, A–C = 15% → the order is A–B–C, because 10 + 5 = 15.

7. Polyploidy	
Definition	The occurrence of one or more **extra sets of all chromosomes** in an organism. A triploid organism is 3n (three complete sets).
In agriculture	Many crops are polyploid: wheat (6n), oats (6n) and sugar cane (8n). Polyploid plants often have increased vigor and size.
Link to Mendel's laws	Gene linkage and polyploidy do not follow all of Mendel's laws of inheritance.

Part 2: Practice Questions

Question	1	
The passing of traits from one generation to the next is called:		

A	mutation	**B**	recombination
C	inheritance (heredity)	**D**	polyploidy

Question	2	
Plants that always produce offspring with the same trait when they self-fertilize are called:		

A	true-breeding	**B**	hybrid
C	heterozygous	**D**	recombinant

Question	3	
In cross-pollination, the male and female gametes come from:		

A	the same flower	**B**	the same plant
C	the same anther	**D**	different plants

Question	4	
What is the genotype of a plant that is homozygous recessive for seed color (y = green)?		

A	**YY**	**B**	**yy**
C	**Yy**	**D**	**Y**

Question	5	

In the F_1 generation of a cross between pure yellow (YY) and pure green (yy) peas, what are the genotype and phenotype?

A	YY, yellow	**B**	yy, green
C	Yy, yellow	**D**	Yy, green

Question	6	

Which statement describes a dominant allele?

A	It controls the trait that appears in the F_1 generation	**B**	It is masked in the F_1 generation
C	It is written as a lowercase letter	**D**	It always produces the more common phenotype in nature

Question	7	

During gamete formation, a plant with the genotype Rr produces:

A	only R gametes	**B**	only r gametes
C	only Rr gametes	**D**	R gametes and r gametes in equal numbers

Question	8	

Which cross is expected to give a phenotypic ratio of 3:1?

A	**TT × tt**	**B**	**Tt × Tt**
C	**Tt × tt**	**D**	**tt × tt**

Question	9	

How many different types of gametes can a plant with the genotype YyRr produce?

A	**1**	**B**	**2**
C	**4**	**D**	**16**

Question	10	

Two genes are very close together on the same chromosome. Which prediction is most reasonable?

A	They will usually be inherited together	**B**	They will sort independently of each other
C	Crossing over will separate them very often	**D**	They must be on different chromosomes

Question	11	

Two linked genes show 7% crossing over between them. How many map units apart are they?

A	**0.7**	**B**	**3.5**
C	**70**	**D**	**7**

Question	12	

A pea plant has 7 pairs of chromosomes. How many different gamete combinations are possible by independent assortment?

A	**14**	**B**	**128**
C	**49**	**D**	**256**

Question	13	

Wheat (6n) is polyploid. What does polyploid mean?

A	It is missing one chromosome	**B**	It lacks one complete set of chromosomes
C	It has one or more extra sets of chromosomes	**D**	Its genes are linked

Questions 14–18: Fill in the blanks using the word bank

Word bank: dihybrid — genotype — linked — phenotype — segregation

14	An organism's allele pair, such as Yy, is called its _____.
15	The observable characteristic of an organism, such as yellow seeds, is called its _____.
16	Genes located close together on the same chromosome are said to be _____.
17	The law of _____ states that the two alleles for each trait separate during meiosis.
18	A cross that follows two traits at the same time is called a _____ cross.

Written Questions

Answer in the space provided (Q19–20)

Question	19	
In pea plants, tall (T) is dominant to dwarf (t). A heterozygous tall plant is crossed with a dwarf plant:		
(a) Write the genotypes of the two parents. **(b)** Construct a Punnett square for the cross. **(c)** State the genotypic ratio and the phenotypic ratio of the offspring. **(d)** State the probability that an offspring will be tall.		
Answer:		

Question	20	
Compare genetic recombination and gene linkage:		
(a) Define each term. **(b)** Explain why linked genes are an exception to the law of independent assortment. **(c)** State how the crossing-over frequency between two genes is related to the distance between them.		
Answer:		

Good luck…